Forced Alignment Using Montreal Forced Aligner (MFA)

Applicant: Syed Khaja Fareeduddin
Project: Speech-to-Speech Machine Translation Internship – Assignment 1
Date: 05-11-2025

1. Objective

The aim of this assignment was to perform forced alignment between audio recordings and their corresponding transcripts using **Montreal Forced Aligner (MFA)**. Forced alignment identifies the start and end times of words and phonemes in speech given a transcript.

2. Methodology

1. Environment Setup
    a. Created a Conda environment named aligner.
    b. Installed Montreal Forced Aligner via Conda.

2. Data Preparation
    a. Used the provided dataset of .wav audio files and .txt transcripts.
    b. Dataset organized according to MFA's required format.

3. Model and Dictionary
    a. Downloaded pretrained acoustic model: english_us_arpa.zip
    b. Downloaded pretrained dictionary: english_us_arpa.dict

4. Forced Alignment Execution
    a. Ran the alignment using the command:
    mfa align --clean --overwrite "dataset" "english_us_arpa.dict" "english_us_arpa.zip" "outputs"
    b. Output: TextGrid files marking word and phoneme boundaries.

5. Output Inspection
    a. Opened TextGrid files in **Praat** software.
    b. Observed phoneme and word boundaries.
    c. Took screenshots from Praat to visualize alignments.

3. Sample Alignment

File: ISLE_SESS0131_BLOCKD02_01_sprt1.TextGrid

Words:
0.0 – 0.44 (silence)
0.44 – 0.53 i
0.53 – 0.92 said
0.92 – 1.33 white
1.33 – 1.48 (silence)
1.48 – 1.8 not
1.8 – 2.25 bait
2.25 – 4.125 (silence)

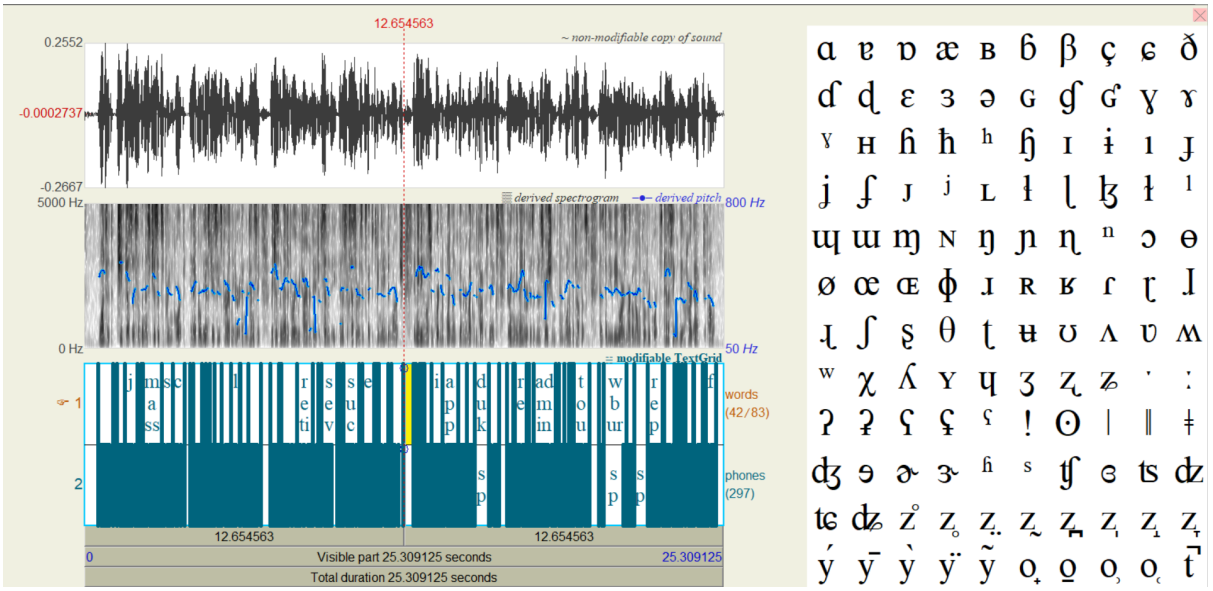
Phonemes:
0.0 – 0.44 (silence)
0.44 – 0.53 AY1
0.53 – 0.71 S
0.71 – 0.79 EH1
0.79 – 0.92 D
0.92 – 1.03 W
1.03 – 1.17 AY1
1.17 – 1.33 T
1.33 – 1.48 (silence)
1.48 – 1.55 N
1.55 – 1.62 AA1
1.62 – 1.8 T
1.8 – 1.84 B
1.84 – 2.05 EY1
2.05 – 2.25 T
2.25 – 4.125 (silence)

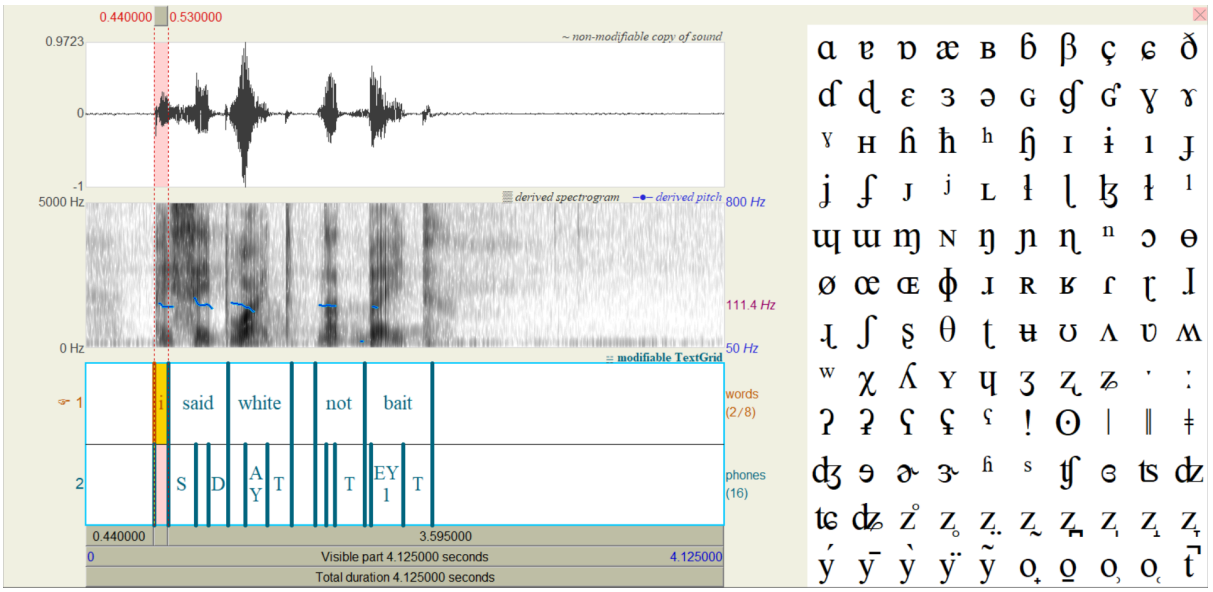
4. Observations

1. MFA successfully aligned all audio files with high accuracy.
2. Minor warnings appeared due to processing a single speaker, but alignment quality remained unaffected.
3. Praat allowed clear visualization of word and phoneme boundaries.

5. Screenshots from Praat

Screenshots of TextGrid visualizations were captured using Praat.
These screenshots illustrate how the word and phoneme intervals are aligned with the audio waveform.
Screenshots are also included in the GitHub repository under the folder: praat_screenshots.





6. Conclusion

The assignment demonstrates how MFA can be effectively used for forced alignment in speech-to-text tasks. Using pretrained models and dictionaries simplifies alignment and allows easy visualization of phonetic and word-level timing information.